

Changes in the Corticospinal Tract Beyond the Ischemic Lesion Following Acute Hemispheric Stroke: A Diffusion Kurtosis Imaging Study

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Background: The degeneration of the corticospinal tract (CST) in chronic stroke has been widely described using diffusion tensor imaging and correlates with the extent of motor deficits. However, only a few studies have reported the early degeneration in the distal CST during the acute stage of stroke and pathological changes in the distal CST have not been described.

Purpose: To study the microstructural changes along the CST beyond the ischemic lesion in acute stroke using diffusion kurtosis imaging (DKI).

Study Type: Prospective.

Population: In all, 48 patients (26 males, 22 females; mean age 58.27 ± 12.89 years) with acute ischemic stroke.

Sequence: A DKI sequence with three b-values (0, 1000, and 2000 s/mm²) at 3.0T MRI.

Assessment: The kurtosis and tensor parameters were derived from DKI and were compared along the length of the CST beyond the ischemic lesion between the affected and unaffected hemispheres using both voxelwise and slice-wise analysis. The degree of neurological deficits was evaluated using the National Institute of Health Stroke Score (NIHSS) and the Barthel index and the clinical outcome at 3 months was evaluated using a modified Rankin scale.

Statistical Tests: Paired t-tests, a linear mixed model, and multivariate linear regression.

Results: Voxelwise analysis demonstrated increased mean kurtosis, increased axial kurtosis, and decreased axial diffusivity in the affected CST, which were seen only at the level of the cerebral peduncle (all corrected $P < 0.05$). Slice-wise analysis also demonstrated increased axial kurtosis in the cerebral peduncle of the affected CST (corrected $P < 0.05$). The axial kurtosis from slice-wise analysis independently correlated with the motor component of NIHSS ($\beta = 0.297$, $P = 0.040$).

Data Conclusion: Our findings suggest that early anterograde degeneration occurs along the axon direction in the distal CST in acute stroke, and can be detected using DKI. Moreover, acute axonal degeneration along the CST correlated with motor deficits.

Level of Evidence: 2

Technical Efficacy Stage: 1

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THE CORTICOSPINAL TRACT (CST) is formed by large pyramidal neurons originating from the motor cortex. These travel through the corona radiata, posterior limb of the internal capsule, cerebral peduncle, and terminate in the spinal cord.¹ The integrity of the CST is essential for stroke recovery from motor deficits,² and can be demonstrated in vivo by diffusion magnetic resonance imaging (MRI) methods, such as diffusion tensor imaging (DTI). The anterograde degeneration of CST at the chronic stage of stroke has been widely investigated in previous studies. Fractional anisotropy (FA) has been the most frequently used parameter for the detection of CST damage. Many studies have demonstrated that reduced FA in the descending CST in chronic stroke correlates with poor motor recovery.^{3–8} However, imaging at later timepoints has no significant impact on the treatment decision because the late stage of CST degeneration is irreversible.

A small number of studies have demonstrated imaging evidence of early degeneration in the distal CST during the acute stage of stroke. One study reported that 25% of a small sample of 20 adult patients had decreased apparent diffusion coefficient (ADC) at the level of the cerebral peduncle 7 days after stroke onset.⁹ Another study demonstrated FA reductions in the CST near the ischemic lesion within a mean of 3 days after onset.¹⁰ The reason that identification of CST degeneration in acute stroke has been seldom reported may indicate that the microstructural changes are too subtle to be detected using conventional approaches and that a diffusion technique offering complementary information and an accurate method of tract analysis are required.

Diffusional kurtosis imaging (DKI), an extension of the traditional DTI model, characterizes non-Gaussian diffusional properties and is more comprehensive in evaluating microstructural complexity in white matter.^{11,12} The growth of publications in brain tumors,^{13,14} neurodegeneration disease,^{15,16} and cerebrovascular disease^{17–19} already suggest that DKI is a promising imaging modality in detecting microstructural changes in human brains not revealed by the Gaussian DTI model.

The definition of 2D or 3D regions of interest (ROIs) in a specific CST region such as the pons is the most common method employed in previous DTI studies^{3–7,9,10}; however, it cannot capture the microstructural integrity of the entire CST. Tractography provides an objective method for estimating the whole CST structural characteristics with minimal operator-dependent bias.²⁰ Statistics analysis along the entire length of the CST can be performed on voxelwise and slice-wise analysis.^{21,22}

In this study we aimed to investigate the microstructural changes along the CST beyond the ischemic lesion in patients with acute anterior circulation infarction using DKI and to assess the relationships between diffusion metrics and neurological functions.

Materials and Methods

Patients

The study was approved by the Medical Ethics Committee of the Second Affiliated Hospital, Zhejiang University School of Medicine. Written informed consent was obtained from all subjects.

Patients with ischemic stroke at the acute phase (3 to 7 days after onset) were recruited prospectively from the Department of Neurology in our institution. Inclusion criteria were as follows: 1) first-ever ischemic stroke; 2) unilateral lesion in the anterior circulation territory; 3) supratentorial CST involved by ischemic lesion; 4) right-hand dominance; 5) age greater than 18 years; 6) without thrombolytic or recanalization therapies; 7) no evidence of hemorrhagic infarction. Exclusion criteria were: i) history of neurological or psychiatric disorders; ii) prestroke modified Rankin scale (mRS) >1.

Neurological deficits before MRI scans were assessed by the National Institute of Health Stroke Score (NIHSS) and the Barthel index (BI). NIHSS is a 15-item stroke scale, with score ranging from 0 to 42 (the higher the score, the greater the function deficits) and BI consists of 10 items that mainly measure daily living activities and mobility, with scores ranging from 0 (totally dependent) to 100 (completely independent).²³ The degree of motor deficits was evaluated using the motor component of NIHSS (m-NIHSS). Clinical outcome at 3 months was evaluated by mRS, which is an ordered scale coded from 0 (no symptoms at all) through 5 (severe disability) or 6 (death).²³ An experienced stroke neurologist with 10 years of experience who was blinded to the imaging findings assessed clinical scores.

MRI Protocol

All subjects were imaged using a 3.0T whole body-scanner (Discovery MR750, GE Healthcare, Milwaukee, WI) with an eight-channel phased array head coil. Before DKI scanning, conventional MR sequences including DWI (relaxation time / echo time [TR/TE] = 5000/88 msec, flip angle = 90°, slice thickness = 4 mm, matrix = 256 × 256, field of view [FOV] = 24 cm, b value = 1000 s/mm² along three orthogonal directions), T₂-FLAIR (TR/TE = 7500/152 msec, inversion time [TI] = 2100 msec, flip angle = 90°, slice thickness = 4 mm, matrix size = 320 × 256, FOV = 24 cm) and T₁-FLAIR (TR/TE = 1750/25 msec, TI = 780 msec, flip angle = 90°, slice thickness = 4 mm, matrix size = 320 × 256, FOV = 24 cm) were obtained, in order to exclude intracranial tumor and hemorrhage. DKI was acquired using a spin-echo echo-planar imaging (EPI) sequence with 30 noncollinear diffusion directions and three b values of 0, 1000, and 2000 sec/mm² and with following parameters: TR/TE = 5000/95 msec, flip angle = 90°, slice thickness = 4 mm, matrix = 256 × 256, FOV = 24 cm.

Imaging Processing, Analysis and Statistics

LESION MASK. A binary lesion mask was used to exclude the effect of lesions on the along-tract analysis of CST. First, the individual lesion was segmented on DWI images using a semiautomated threshold algorithm to identify areas with an intensity greater than the mean +2 standard deviations of the value in the homologous contralateral region.²⁴ Second, all right-sided lesions were flipped

to the left side so that all the lesions were in the left hemisphere. Third, all lesions were normalized to the Montreal Neurological Institute (MNI) space and resliced with a voxel size of 1 mm^3 and were combined to create a lesion overlap map (Fig. 1). Finally, the lesion mask was binarized based on the lesion overlap map.

PROCESSING OF KURTOSIS TENSOR AND DIFFUSION TENSOR METRICS. Each scan of diffusion-weighted images was preprocessed using FMRIB's Software Library (FSL, v. 6.0, Oxford, UK), including correcting for signal distortions due to eddy currents and head motion, rotating the gradient directions according to eddy currents correction and removing nonbrain tissue.

Diffusional kurtosis (K) was computed by fitting all diffusion-weighted images to the DKI model¹¹:

$$\ln[S(b)/S(0)] = -bD + \frac{1}{6}Db^2K^2 \quad (1)$$

where $S(b)$ is the diffusion-weighted signal intensity as a function of b -value, $S(0)$ is the signal intensity at $b = 0 \text{ sec/mm}^2$, and D is the apparent diffusion coefficient. Similar to the diffusion tensor model, a rank 4 tensor, $W_{[3 \times 3 \times 3 \times 3]}$, is defined as the diffusion kurtosis tensor with 15 independent components (since it is fully symmetric).¹¹ Therefore, the general form for Eq. (1) can be written as follows:

$$\begin{aligned} \ln[S(b, n)] = & \ln[S(0)] - b \sum_{i=1}^3 \sum_{j=1}^3 n_i n_j D_{ij}^{app} \\ & + \frac{1}{6} b^2 \left(\frac{1}{3} \sum_{i=1}^3 D_{ii}^{app} \right)^2 \\ & \times \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \sum_{l=1}^3 n_i n_j n_k n_l W_{ijkl}^{app} + O(b^3) \end{aligned} \quad (2)$$

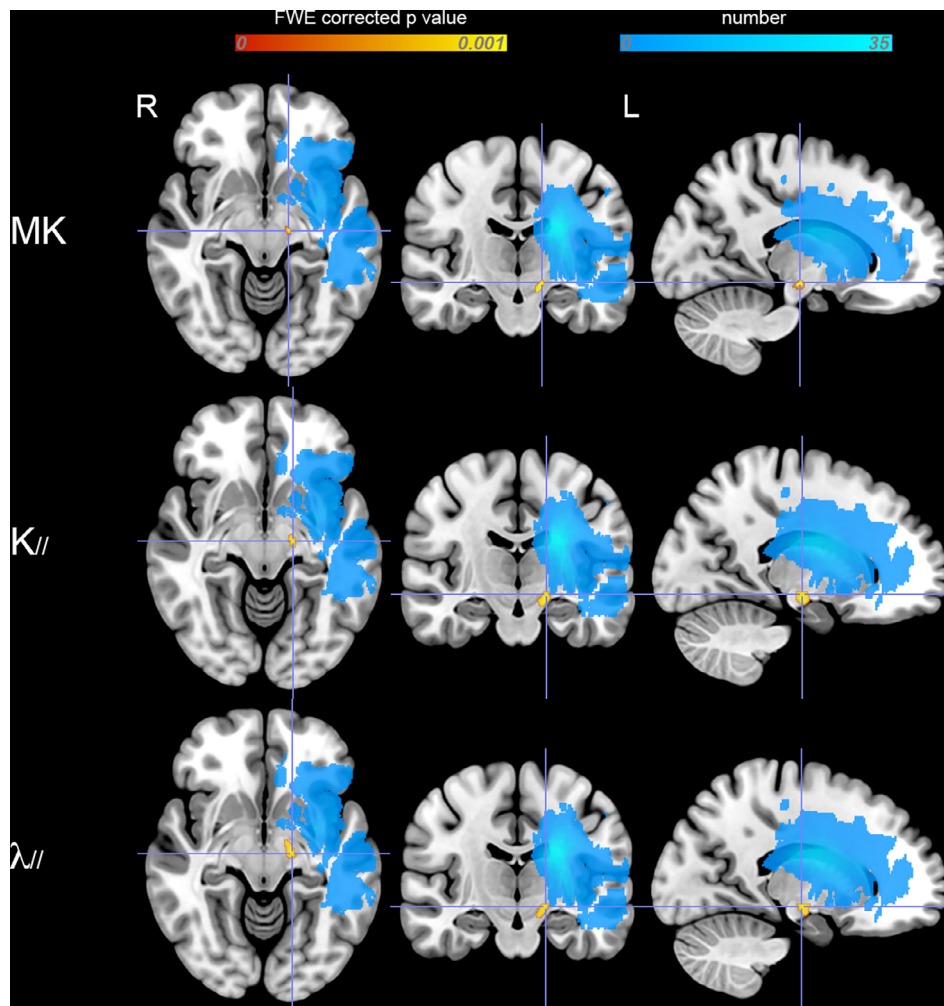


FIGURE 1: Lesion overlap map and voxelwise results along the corticospinal tract (CST) overlaid on the standard Montreal Neurological Institute (MNI) template. All lesions were flipped onto the left hemisphere. The right color bar represents the number patients with a lesion to a specific voxel. Increased mean kurtosis (MK) with a cluster size of 130 and familywise error (FWE) corrected P value of 0.027, axial kurtosis ($K_{||}$) with a cluster size of 277 and FWE corrected P value of 0.012, and decreased axial kurtosis ($\lambda_{||}$) with a cluster size of 554 and FWE corrected P value of 0.004 were detected in the regions of cerebral peduncle of the affected CST, compared with the unaffected CST. The left color bar represents the range of FWE corrected P value. L represents the left hemisphere and R represents the right hemisphere.

in which D^{app} and W^{app} are the apparent diffusion and kurtosis measures along the direction vector n . This model has 21 unknown independent parameters, 6 parameters for diffusion tensor and 15 parameters for kurtosis tensor, which needed to be estimated from the measured diffusion signal.

We estimated the kurtosis tensors by solving Eq. (2) using least-squares optimization in MatLab R2017b (MathWorks, Natick, MA). Finally, mean kurtosis (MK), axial kurtosis ($K_{||}$), and radial kurtosis ($K_{\perp}$) were calculated.

The preprocessed diffusion images with $b = 0 \text{ sec/mm}^2$ and $b = 1000 \text{ sec/mm}^2$ from DKI data were used for DTI fitting. FA, mean diffusivity (MD), axial diffusivity ($\lambda_{||}$), and radial diffusivity ($\lambda_{\perp}$) were generated using DTIFIT in FSL.

MK, $K_{||}$, $K_{\perp}$, FA, MD, $\lambda_{||}$, and $\lambda_{\perp}$ maps from each subject were normalized into the MNI (Montreal, Canada) template with a voxel size of 1 mm^3 using the Statistical Parametric Mapping Version 12 (SPM12; Wellcome Department of Cognitive Neurology, London, UK).

CORTICOSPINAL TRACT TRACTOGRAPHY. To create CST template, CST tractography was performed in the unaffected hemisphere in each patient using the preprocessed DTI data ($b = 0 \text{ sec/mm}^2$ and $b = 1000 \text{ sec/mm}^2$) by a probabilistic fiber-tracking algorithm implemented in FSL (probtrackx). First, the ROIs of the cerebral peduncle and posterior limb of the internal capsule, which were manually drawn on an FA map by an experienced neuroradiologist with 6 years' experience were used as seed masks. Tractography was performed using the following parameters: 5000 streams per each voxel seed, 0.5 mm step length, and 0.2 curvature threshold. Then CST was normalized into MNI space with a voxel size of 1 mm^3 and binarized at a threshold of 5% of the maximum voxel value to remove improbable pathways. The CST template was averaged by binarized CST from each patient. For the left CST template, CST from the right hemispheres were flipped to the left, and then they were overlapped with CST from the left hemispheres, and finally the group-level overlap of the left CST was binarized at a threshold of 20%. The right CST template was created similarly. Lastly, the masks of the CST template of two hemispheres were used to segment the CST regions from the normalized diffusion maps including MK, $K_{||}$, $K_{\perp}$, FA, MD, $\lambda_{||}$, and $\lambda_{\perp}$. Additionally, the volume of the CST lesion load was measured.

VOXELWISE STATISTICAL ANALYSIS. For voxelwise analysis, in order to perform group analysis, the affected CST and unaffected CST from each diffusion map at the right hemisphere were flipped to the left so that all CSTs were transformed to the same hemisphere. Then all the diffusion maps of CST were smoothed using a Gaussian filter with full-width at half-maximum of 4 mm. The significant differences in diffusion parameters between the affected CST and unaffected CST were performed using paired t -tests with age as a covariate in SPM12. The significant results were masked by the binary lesion mask and only differences in the areas beyond the lesion were reported at $P < 0.05$ with a minimal cluster size of 20 voxels after familywise error (FWE) correction for multiple comparisons.

TABLE 1. Demographic and Clinical Characteristics of 48 Patients With Ischemic Stroke

Age, mean $\pm$ SD, y	58.27 $\pm$ 12.89
Men, No. (%)	26 (54.2)
Days between stroke onset and MRI scan, mean $\pm$ SD, d	5.44 $\pm$ 1.29
Vascular risk factors	
Hypertension, No. (%)	36 (75.0)
Diabetes, No. (%)	23 (47.9)
Hyperlipidemia, No. (%)	5 (10.4)
Smoking, No. (%)	19 (39.6)
Drinking, No. (%)	15 (31.3)
Left-sided lesion, No. (%)	24 (50.0)
Lesion size, median, IQR, cm^3	6.74 (0.24–76.02)
CST lesion load, median, IQR, cm^3	0.66 (0.14–2.85)
NIHSS score, mean $\pm$ SD	8.04 $\pm$ 3.05
m-NIHSS, mean $\pm$ SD	4.50 $\pm$ 2.54
BI, mean $\pm$ SD	44.79 $\pm$ 23.61
mRS, median, IQR	2 (0–4)

SD: standard deviation; MRI: magnetic resonance imaging; IQR: interquartile range; CST: corticospinal tract; NIHSS: National Institute of Health Stroke Score; m-NIHSS: motor component of the NIHSS; BI: Barthel index; mRS: modified Rankin scale.

SLICEWISE STATISTICAL ANALYSIS. For slice-wise analysis, ROIs for the affected and unaffected CST were cut into transversal slices along the z -axis. The mean values of MK, $K_{||}$, $K_{\perp}$, FA, MD, $\lambda_{||}$, and $\lambda_{\perp}$ were then extracted from each slice of CST and plotted against the z -axis. There were 119 slices of CST (from slice 14 to 132) in our subjects. The significant differences in diffusion parameters in each slice of CST between two hemispheres were compared by a linear mixed model using SPSS software v. 20.0 for Windows (IBM, Armonk, NY). Age was included as a covariate. Statistical significance was accepted at $P < 0.05$. A Bonferroni-corrected $P < 0.05$ for multiple comparisons was used in post-hoc analysis for DKI ($P < 0.05/[119 \text{ slices} \times 3 \text{ metrics}] = 0.00014$) and DTI ($P < 0.05/[119 \text{ slices} \times 4 \text{ metrics}] = 0.00011$). Uncorrected $P < 0.05$ was considered a trend in post-hoc analysis. Only slices beyond the ischemic lesions (based on the binary lesion mask, corresponding to the slices from 65 to 121) were analyzed.

ASSOCIATION WITH CLINICAL SCORES. The mean values of diffusion parameters were extracted from the significant clusters in voxelwise analysis and from the significant slices in slice-wise analysis. The percent change (Δ) of diffusion parameters between two sides

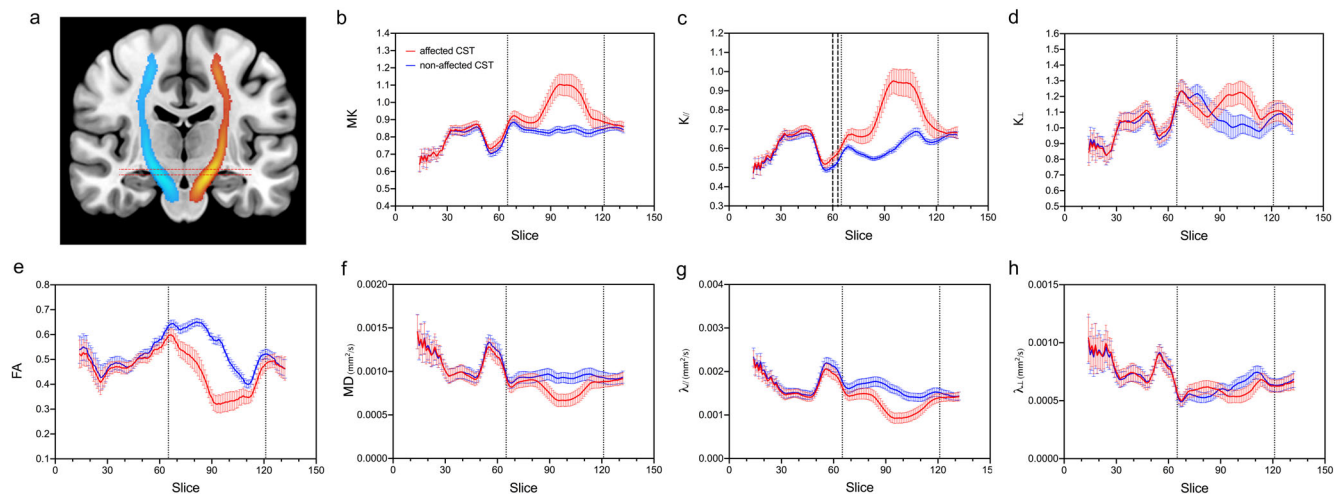


FIGURE 2: Slicewise paired *t*-tests along the corticospinal tract (CST). (a) The affected CST (red) and unaffected CST (blue) were shown in the coronal plane. (b–h) The results of mean kurtosis (MK), axial kurtosis ($K_{||}$) and radial kurtosis ($K_{\perp}$), fractional anisotropy (FA), mean diffusivity (MD), axial diffusivity ($\lambda_{||}$), and radial diffusivity ($\lambda_{\perp}$) were plotted. Among the diffusion parameters, only significantly increased $K_{||}$ was found at the slices (from slice 60–63, the dashed line) of cerebral peduncle of the affected CST, corresponding to the regions between red dashed lines in (a). The dotted line indicates the slices of ischemic lesions. Each point in the curve represents the mean value of the data with the error bars.

of CST were further calculated by (Δ diffusion parameter = [diffusion parameter at the affected CST - diffusion parameter at the unaffected CST] / diffusion parameter at the unaffected CST). The relationships between Δ diffusion parameters with clinical information were analyzed using Pearson's or Spearman's correlation coefficients according to data normality. Multivariate linear regression with the stepwise method was further used to analyze the independent factor related to clinical scores, and the standardized coefficient (β) of factors was estimated. A two-tailed value of $P < 0.05$ was considered significant.

Results

Patient Characteristics

A total of 48 patients were included in this study. The mean patient age was 58.27 ± 12.89 years (range, 28–81 years) with 26/48 (54.2%) male. The median volume of ischemic lesion was 6.74 cm^3 (range, $0.24\text{--}76.02 \text{ cm}^3$) and CST lesion load was 0.66 cm^3 (range, $0.14\text{--}2.85 \text{ cm}^3$). The time

between stroke onset and MRI scan was 5.44 ± 1.29 days (range, 3–7 days). The NIHSS score was 8.04 ± 3.05 (range, 4–14) and the m-NIHSS score was 4.50 ± 2.54 (range, 0–8). BI was 44.79 ± 23.61 (range, 0–100). The median score of mRS was 2 (range, 0–4). Table 1 summarizes the baseline patient characteristics.

Voxelwise Analysis of CST

Voxelwise analysis revealed increased MK (coordinates of peak voxel, $-13, -19, -11$; FWE corrected $p_{\text{peak-level}} = 0.009$; FWE corrected $p_{\text{cluster-level}} = 0.027$; cluster size = 130), increased $K_{||}$ (coordinates of peak voxel, $-16, -15, -13$; FWE corrected $p_{\text{peak-level}} = 0.001$; FWE corrected $p_{\text{cluster-level}} = 0.012$; cluster size = 277) and decreased $\lambda_{||}$ (coordinates of peak voxel, $-11, -22, -18$; FWE corrected $p_{\text{peak-level}} = 0.001$; FWE corrected $p_{\text{cluster-level}} = 0.004$; cluster size = 554) at the level of the cerebral peduncle of the affected CST. The results are

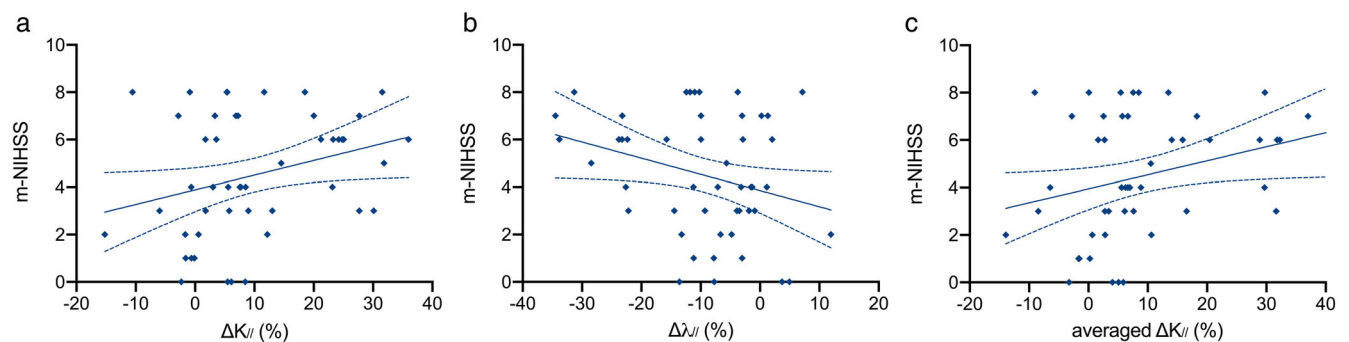


FIGURE 3: Scatter dot plots (a–c) show significant correlations between percent change of axial kurtosis ($\Delta K_{||}$), axial diffusivity ($\Delta \lambda_{||}$) from the voxel-based result, and averaged $\Delta K_{||}$ from sliced-based results with a motor component of the National Institute of Health Stroke Score (m-NIHSS).

shown in Fig. 1. No significant differences in $K_{\perp}$, FA, MD, and $\lambda_{\perp}$ were found.

Slicewise Analysis of CST

A linear mixed model showed $K_{\parallel}$ ($P < 0.001$) and MK ($P < 0.001$) along the CST had significant differences between two hemispheres. In post-hoc analysis, increased $K_{\parallel}$ was only found at the slices from 60 to 63 (all corrected $P < 0.05$), corresponding to the cerebral peduncle of the affected CST, after multiple comparison correction. However, no significant slices were found in MK after multiple comparison correction. We also found a trend towards increased $K_{\parallel}$ in three slices (slice 122, uncorrected $P = 0.012$; slice 123, uncorrected $P = 0.017$; slice 124, uncorrected $P = 0.033$) of CST above the lesion. The results are shown in Fig. 2.

Correlation With Clinical Characteristics

ΔMK ($r = 0.386$, $P = 0.007$; $r = 0.431$, $P = 0.002$), $\Delta K_{\parallel}$ ($r = 0.307$, $P = 0.034$; $r = 0.372$, $P = 0.009$) and $\Delta \lambda_{\parallel}$ ($r = -0.315$, $P = 0.029$; $r = -0.516$, $P < 0.001$) from voxel-based analysis were correlated with CST lesion load and time after stroke onset. The averaged $\Delta K_{\parallel}$ ($r = 0.338$, $P = 0.019$; $r = 0.452$, $P = 0.001$) from slice 60 to 63 of CST was also correlated with CST lesion load and time after stroke onset. No significant correlations were found between diffusion parameters and age, gender, or lesion size. m-NIHSS significantly correlated with $\Delta K_{\parallel}$ ($r = 0.294$, $P = 0.043$) and $\Delta \lambda_{\parallel}$ ($r = -0.288$, $P = 0.047$) from voxel-based analysis and averaged $\Delta K_{\parallel}$ ($r = 0.297$, $P = 0.040$) from sliced-based analysis (Fig. 3a-c), whereas NIHSS, BI, and 3-month mRS did not correlate with these parameters. The multivariate linear model showed that only averaged $\Delta K_{\parallel}$ independently correlated with m-NIHSS ($\beta = 0.297$, $P = 0.040$), after correcting for other variables including age, gender, CST lesion load, and time after stroke.

Discussion

In this study we described abnormal increased MK, $K_{\parallel}$, and decreased $\lambda_{\parallel}$ in the regions of the cerebral peduncle of the affected CST, compared with the unaffected CST in the acute stage of anterior circulation stroke. Increases in MK, $K_{\parallel}$, and a decrease in $\lambda_{\parallel}$ in the distal CST correlated with CST lesion load, time after stroke onset, and an increase in $K_{\parallel}$ independently correlated with the severity of motor deficits.

Although degeneration of distal efferent CST fibers in chronic stroke is well recognized, secondary injury in descending CST in the acute stage of stroke has rarely been reported. In this study, increased MK and $K_{\parallel}$ and decreased $\lambda_{\parallel}$ were demonstrated in the cerebral peduncle within the first week after infarction. Increased $K_{\parallel}$ and decreased $\lambda_{\parallel}$ indicate that water diffusion is restricted along the axonal direction.²⁵

The increased fiber dispersion in the cerebral peduncle of the ipsilesional hemisphere detected by neurite orientation dispersion and density imaging has been reported in a recent study,²⁶ further supporting our findings. After an acute ischemic event the axonal segments distal to the injury exhibit increased Ca^{2+} influx, which triggers acute axon degeneration.^{27,28} In murine models the delay between injury and axon degeneration is 24 to 48 hours, whereas in humans it takes several days.²⁹ Eventually, axon pathology is characterized by beading and swelling before catastrophic granular disintegration of the cytoskeleton occurs.³⁰ Therefore, diffusion changes in the axonal direction in our study may represent the early process of axonal beading and swelling.

In addition to anterograde degeneration of CST, retrograde degeneration along the CST also occurs following ischemic stroke. Retrograde CST degeneration, known as the dying-back phenomenon, is hypothesized to represent retrograde degeneration beginning at the distal ends of damaged axons.³¹ However, only a few human studies have reported the imaging evidence of retrograde degeneration in the CST after stroke and, in these cases, microstructural degeneration detected by DTI was observed at least 1 week after onset.^{32–34} Although our study did not find significant diffusion changes in the proximal portion of CST beyond lesions, slicewise analysis demonstrated that a trend towards higher $K_{\parallel}$ in the slices above the lesion. This finding suggests that retrograde degeneration may be stable or evolve slowly during the acute stage of stroke.

Our study showed that the extent of diffusion restriction in the axonal direction in the cerebral peduncle was an independent factor of motor deficits after stroke, which is in agreement with previous findings that microstructural injury in the descending CST correlated well with motor impairment in patients with chronic stroke.^{3–8} We further found that early degeneration of the distal CST depended on the size of the CST lesion load and the time after stroke. At the present time, access to reperfusion therapy is limited by the narrow therapeutic window. Novel therapeutic strategies are clearly required and extensive research is ongoing. One recent experimental study demonstrated that removal of extracellular calcium with a Ca^{2+} chelator or pharmacologically inhibiting calpain can delay or prevent acute axon degeneration.³⁵ Early CST degeneration has the potential to provide a new biomarker of early degenerative change for use in clinical trials and possibly in patient selection for this and similar novel therapeutic strategies.

The voxelwise analysis and slicewise analysis in our study offer complementary information concerning the pattern and severity of secondary degeneration in the CST. On the one hand, voxel-based analysis provides more changes in diffusion parameters, although both methods reveal that $K_{\parallel}$ shows a significant change in the distal site of CST. In comparison, slice-based analysis reveals a trend to similar

microstructural change in the proximal CST, which is not evident in the voxelwise approach. Compared to the more widely used ROI-based method, both voxelwise and slice-wise analytical approaches can reveal more detailed information about microstructural damage along the entire CST.

The finding of distal degeneration in the cerebral peduncle is compatible with previous studies,^{9,10} although no significant changes in FA or MD were found in our study. The significant difference in MK but not in FA or MD from voxelwise analysis and the significant difference in $K_{||}$ rather than $\lambda_{||}$ from slice-wise analysis both suggest that DKI is more sensitive for detecting early microstructural degeneration than DTI. This supports the observations of recent studies that showed that DKI demonstrates larger changes than DTI in the detection of microstructural changes in ischemic tissues,¹⁷ as well as in the CST¹⁹ after stroke. However, this study is the first to report the comparative sensitivity of DKI and DTI to microstructural changes in the CST beyond the ischemic lesion.

Interestingly, for patients with hemorrhagic stroke, FA derived from DTI is sensitive in detecting early microstructural changes within 2 to 3 days after onset in the cerebral peduncle of the CST.^{36,37} A recent study demonstrated that the ratio of FA between two sides of the cerebral peduncle in patients with hemorrhagic strokes is significantly lower than that in patients with ischemic stroke.³⁸ This may be caused by the different mechanisms of CST degeneration between hemorrhagic stroke and ischemic stroke.

Limitations

There are some limitations in this study. The first is that the patients included had only mild to moderate impairments after stroke. In patients with more severe functional deficits, a change in diffusion metrics may be more pronounced. Second, we use the tracked CST of the unaffected hemisphere to simulate the CST in the affected hemisphere; this approach may induce error in estimating the extent of CST injury. However, it is difficult to accurately track the CST in the affected hemisphere because the ischemic lesion directly affects not only the fibers but the diffusion signal itself. Third, white matter hyperintensities (WMHs) are highly prevalent in older adults aged 60 years and over,³⁹ and they may produce a potential source of variance for the results. Although the distribution of WMHs locates outside the pathway of CST in our study, the effect of normal-appearing white matter surrounding WMHs on the microstructure of CST at the slice such as corona radiata cannot be completely excluded. The fourth limitation is the 4-mm thickness used for the along-tract analysis. A thinner thickness would be more appropriate to measure the microstructural changes of CST, but it largely increases the scanning time with two b values (1000 and 2000 sec/mm²), thus giving rise to the head

motion. Finally, this was a single-center study with a relatively small sample size.

Conclusion

Our study provides the imaging evidence of early anterograde degeneration along the axon direction in the distal CST within 1 week after stroke and the axonal degeneration correlates with the motor deficits. The technique of DKI demonstrates tissue microstructural changes in the CST not identified by FA and MD metrics.

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